

Detecting IoC and Reviewing Threat Intelligence Sources

Introduction

The Detection Lab project was geared towards showcasing my Security Operations skills by detecting IoCs and reviewing threat intelligence sources in my home Lab. Analyzing the IoCs with the exploration of publicly available threat intelligence sources will enable me as a Security Analyst to rapidly evaluate potential threats, understand their context, hunt for potential compromise, and initiate effective response measures.

Indicators of Compromise (IoCs): They are artifacts or pieces of evidence identified within a network or systems that may mirror malicious activity. The most common types of IoCs include:

- **IP addresses:** This may be associated with malicious servers or sources of attacks.
- **File Hashes:** This pertains to the unique identification of files that can help facilitate the detection of malware across several systems.
- **Domain Names / URLs:** These are often used in phishing campaigns or as part malicious command-and control infrastructure.

Cybersecurity analysts and threat hunters gather indicators of compromise (IoCs) to detect, examine, and address potential threats. IoCs indicate that a security incident might have taken place, such as network traffic from an IP address or domain linked to malicious behavior. These indicators can be found in:

- System and application logs,
- Network monitoring tools
- Endpoint protection solutions,
- Security Information and Event Management (SIEM) systems.

Tools

To demonstrate the steps taken for the experiment in my home lab, I utilized the following free tools.

- **VirusTotal:** It is a powerful malware scanning service, created by a Spanish security company, which helps Security Analysts analyze suspicious files and URLs to uncover different types of malware behaviour and malicious content through files, file hashes, and URL. There is disadvantage when sensitive data are uploaded on VirusTotal, given the fact that anyone who pays

for VirusTotal usage gets an access to any uploaded file and can download that file. This is why VirusTotal emphasized clearly that it is not responsible for any data uploaded on its platform.

- **MalShare:** It is platform of a free malware repository, that provides researchers access to samples and malicious feeds used to analyze file hashes as IoCs, identify malicious files and gather threat intelligence about malware.
- **AbuseIPDB:** It is a platform that is used to review IP addresses as IoCs, which helps provide information on malicious activity, abuse reports and reputation for threat analysis.
- **Cisco Talos Intelligence:** This platform is used is used to investigate domains or URLs as Iocs, evaluating its reputation, related Ips, and security alerts to support threat intelligence.

PART A: Reviewing a File Hash with MalShare / VirusTotal

During this project, MalShare website was shut down after losing their host provider, which made me switch to VirusTotal.

Step 1

- I double-clicked Google Chrome  in my desktop for opening and typed in: <https://virustotal.com/> in the address bar.
- In the Quick Search bar, I typed in a hash value of **44d88612fea8a8f36de82e1278abb02f**, which I obtained for analysis and pressed Enter.

EICAR

- This hash value is extracted from virus, named **EICAR**. It is test string used for detecting and testing antivirus software. It is “dummy virus” that provokes an Antivirus engine trigger an alert.
- This helps Antivirus engines to demonstrate their ability to recognize signature-based malwares and neutralize their threats.

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f					Sign in	Sign up
AVG	① EICAR Test-NOT Virus!!!	Avira (no cloud)	① Eicar-Test-Signature			
Baidu	① Win32.Test.Eicar.a	BitDefender	① EICAR-Test-File (not A Virus)			
Bkav Pro	① W32.EicarTest.Trojan	ClamAV	① Eicar-Test-Signature			
CMC	① Eicar.test.file	CTX	① Txt.virus.eicar			
Cynet	① Malicious (score: 99)	DrWeb	① EICAR Test File (NOT A Virus!)			
Elastic	① Eicar	Emsisoft	① EICAR-Test-File (A)			
eScan	① EICAR-Test-File	ESET-NOD32	① Eicar Test File			
Fortinet	① EICAR_TEST_FILE	GData	① EICAR_TEST_FILE			
Google	① Detected	Gridinsoft (no cloud)	① Trojan.U.EICAR_Test_File.dd			
Huorong	① TEST/AVEngTestFile/EICAR	Ikarus	① EICAR-Test-File			
Jiangmin	① EICAR-Test-File	K7AntiVirus	① EICAR_Test_File			
K7GW	① EICAR_Test_File	Kaspersky	① EICAR-Test-File			
Kingsoft	① Test.eicar.aa	Lionic	① Test.File.EICAR.y			
Malwarebytes	① EICAR-AV-Test	MaxSecure	① VIRUS.EICAR.TEST			
McAfee Scanner	① WPS/EICAR	Microsoft	① Virus:DOS/EICAR_Test_File			
NANO-Antivirus	① Marker.Dos.EICAR-Test-File.dyb	Panda	① EICAR-AV-TEST-FILE			
QuickHeal	① EICAR.TestFile	Rising	① Virus.EICARTestFile!1.103DB (CLASSIC)			

Figure 3: EICAR Virus

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f					Sign in	Sign up
Sangfor Engine Zero	① EICAR-Test-File (not A Virus)	SecureAge	① EICAR Anti-Virus Test File			
SentinelOne (Static ML)	① Static AI - Malicious COM	Skyhigh (SWG)	① EICAR Test File			
Sophos	① EICAR-AV-Test	SUPERAntiSpyware	① NotAThreat.EICAR[TestFile]			
Symantec Mobile Insight	① ALG:EICAR Test String	TACHYON	① EICAR-Test-File			
Tencent	① EICAR.TEST.NOT-A-VIRUS	Trellix ENS	① EICAR Test File			
TrendMicro	① Eicar_test_file	TrendMicro-HouseCall	① Eicar_test_file			
Varist	① EICAR_Test_File	VBA32	① EICAR-Test-File			
VIPRE	① EICAR-Test-File (not A Virus)	VirIT	① EICAR-Test-File			
ViRobot	① EICAR-test	Webroot	① W32.Eicar.Testvirus.Gen			
WithSecure	① EICAR_Test_File	Xcitium	① Malware@#2975xfk8s2pq1			
Yandex	① EICAR_test_file	Zillya	① EICAR.TestFile			
ZoneAlarm by Check Point	① EICAR-AV-Test	Zoner	① EICAR.Test.File-NoVirus.250			
Acronis (Static ML)	✔ Undetected	CrowdStrike Falcon	✔ Undetected			
Symantec	🔒 Timeout	Arctic Wolf	🔒 Unable to process file type			
BitDefenderFalx	🔒 Unable to process file type	DeepInstinct	🔒 Unable to process file type			
Palo Alto Networks	🔒 Unable to process file type	TEHTRIS	🔒 Unable to process file type			
Trapmine	🔒 Unable to process file type	Trustlook	🔒 Unable to process file type			

Figure 4: EICAR Virus

- [illegible]

- In the details section, the history, properties, and the signature names of the EICAR virus are contextualized. This text string was first detected by the offensive security vendors on June 14, 2012. The MD5 hash, SHA-1, SHA-256 sum are depicted as properties of the powershell file. In exceptions of SHA-256, MD5 and SHA-1 are weak algorithms that can allow collision. Weak algorithms like MD5 and SHA-1 constitute cryptographic vulnerabilities, allowing collision attacks, reducing password security. Notable real-world examples are
 - Heartbleed (OpenSSL) - exposed encrypted communications.
 - KRACK (WPA2) - allowed Wi-Fi traffic interception.

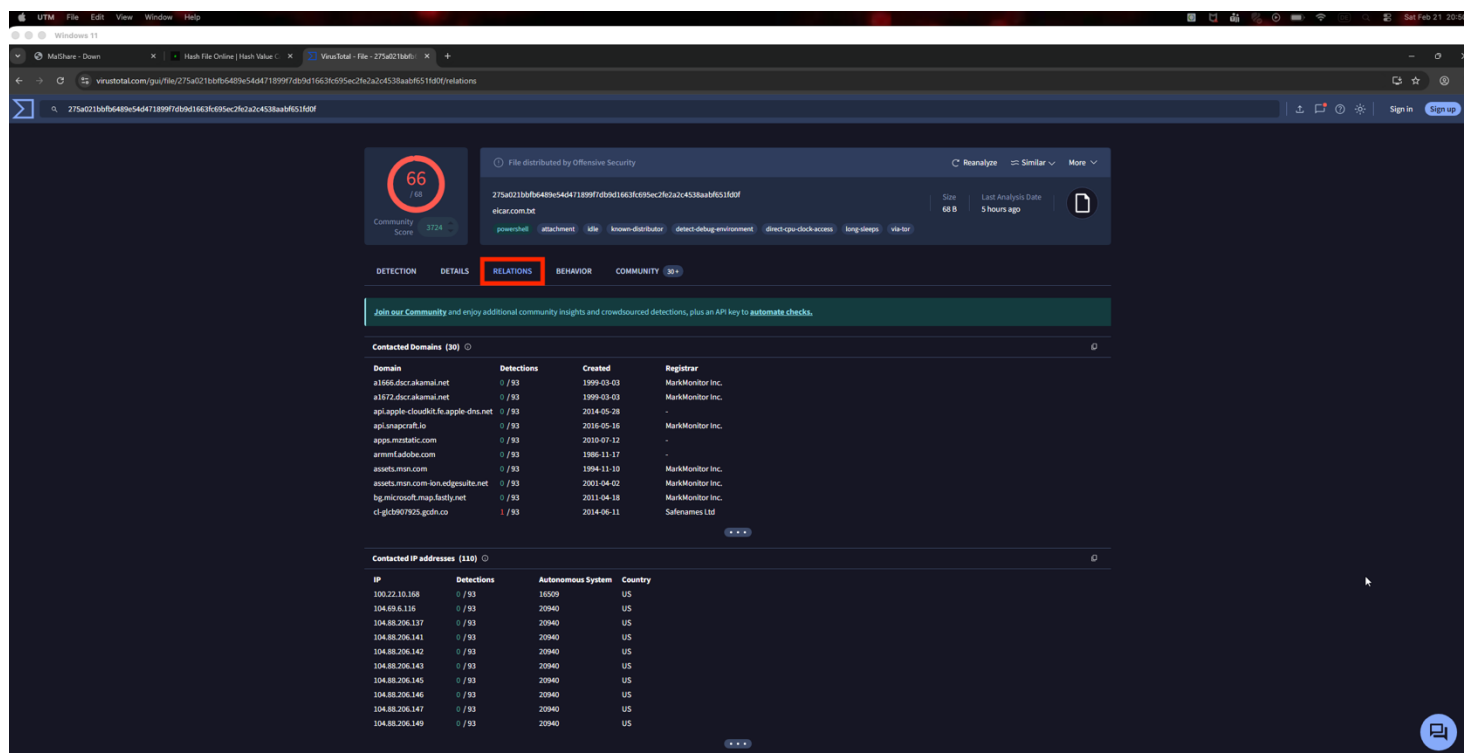


Figure 6.1: Relations Sections

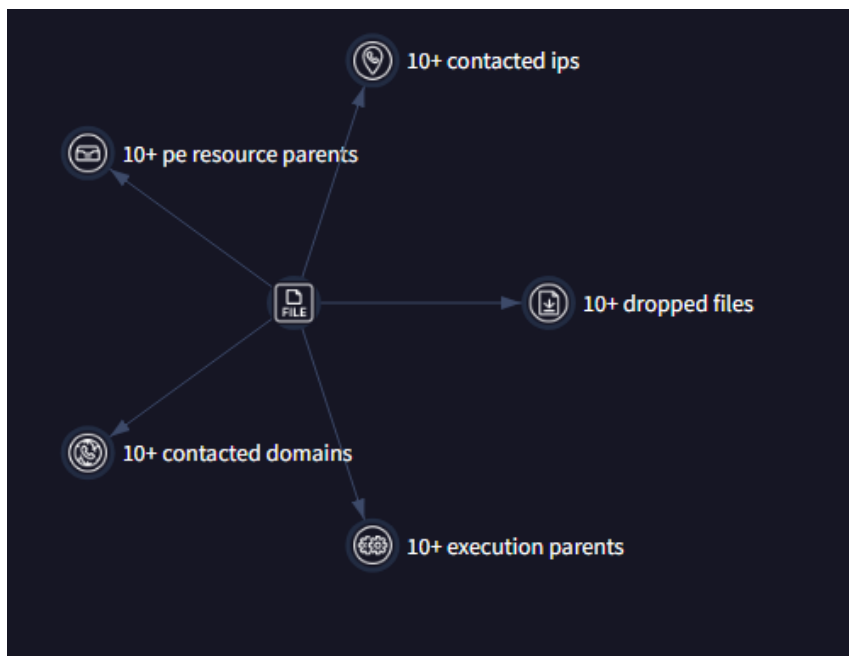


Figure 6.2: Relations Sections

- The Relations Section shows us various kinds of Domains that were contacted by the EICAR virus, the IP addresses that came in contact with the “Dummy Virus,” including Execution parents, and Dropped Files These provide us additional context for investigating this file hash file.

UTMFileEditVirtual MachineViewWindowHelp

MacShare - DownHash File Online | Hash Value :Virtual Total - File - 275a021b6f6489e544718997db9d16636c95ec2f62a2c4538aabf651f0fbehavior

275a021b6f6489e544718997db9d16636c95ec2f62a2c4538aabf651f0f

66/68Community Score3724

File distributed by Offensive Security275a021b6f6489e544718997db9d16636c95ec2f62a2c4538aabf651f0fSize68 BLast Analysis Date5 hours ago

powershellattachmentfileknown distributordetect debug environmentdirect cpu-disk accesslong sleepsva-tor

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY36+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Display grouped sandbox reports

CAPE Linux

CAPE Sandbox

Lastline

OS X Sandbox

VirusTotal Jupyterbox

VirusTotal Observer

Yomi Hunter

Zenbox

Activity SummaryDownload ArtifactsFull ReportsHelp

3 Detections3 MALWARE3 TROJAN1 EVADER

Mitre Signatures1 LOW1 MEDIUM

IDS Rules1 LOW

Sigma Rules1 LOW

Dropped Files7 OTHER1 TEXT1 XML

Network comms35 DNS34 IP2 JAR

Behavior Tags

Dynamic Analysis Sandbox Detections

The sandbox Zenbox flags this file as: MALWARE TROJAN

The sandbox Lastline flags this file as: MALWARE TROJAN

The sandbox OS X Sandbox flags this file as: MALWARE TROJAN EVADER

Figure 7: Behavior Section

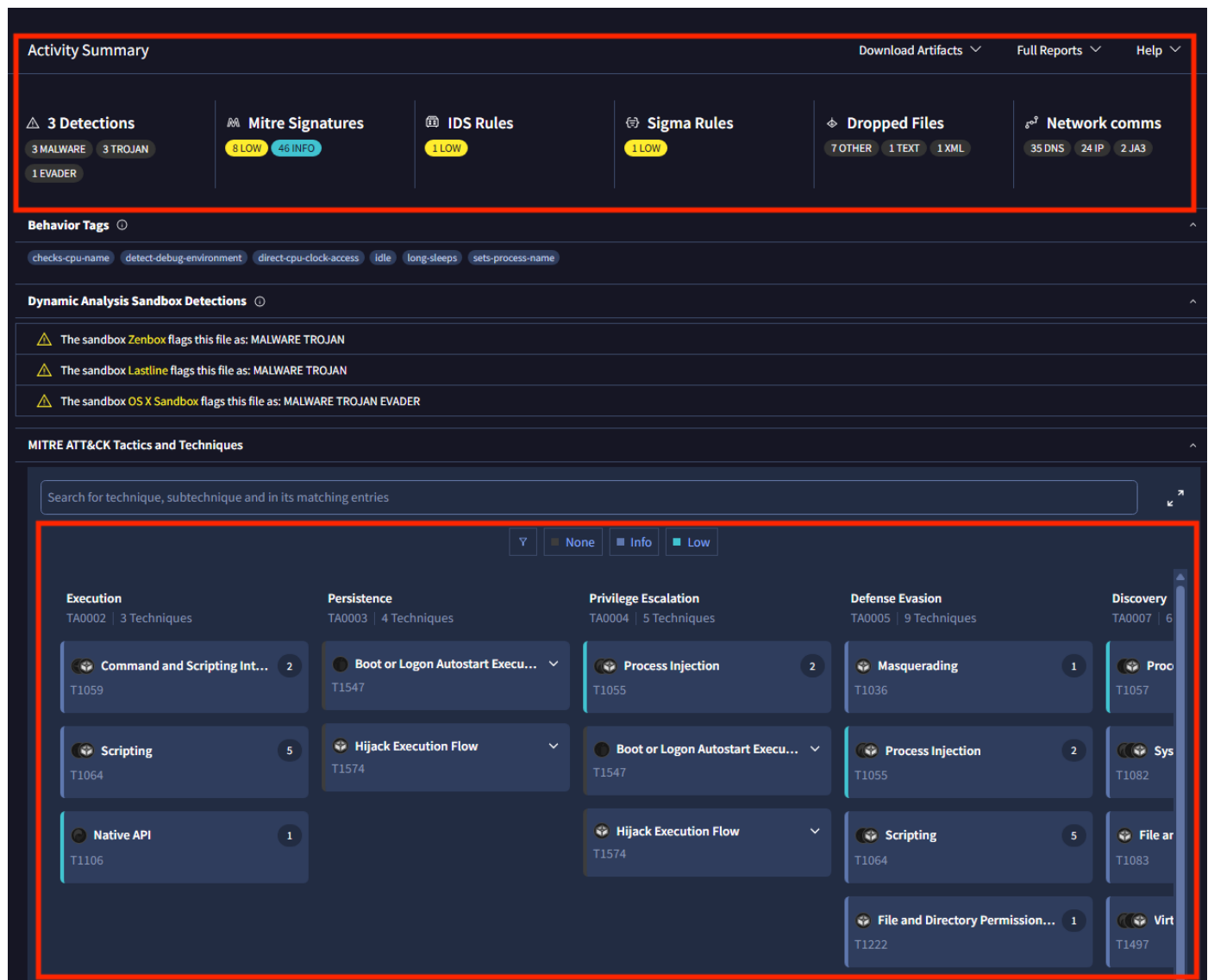


Figure 8: MITRE ATT&CK

The Behavior Section provides us with MITRE Attack tactics and techniques in form of signatures, Intrusion Detections Systems (IDS) Rules, SIGMA Rules, Dropped Files, Network communications.

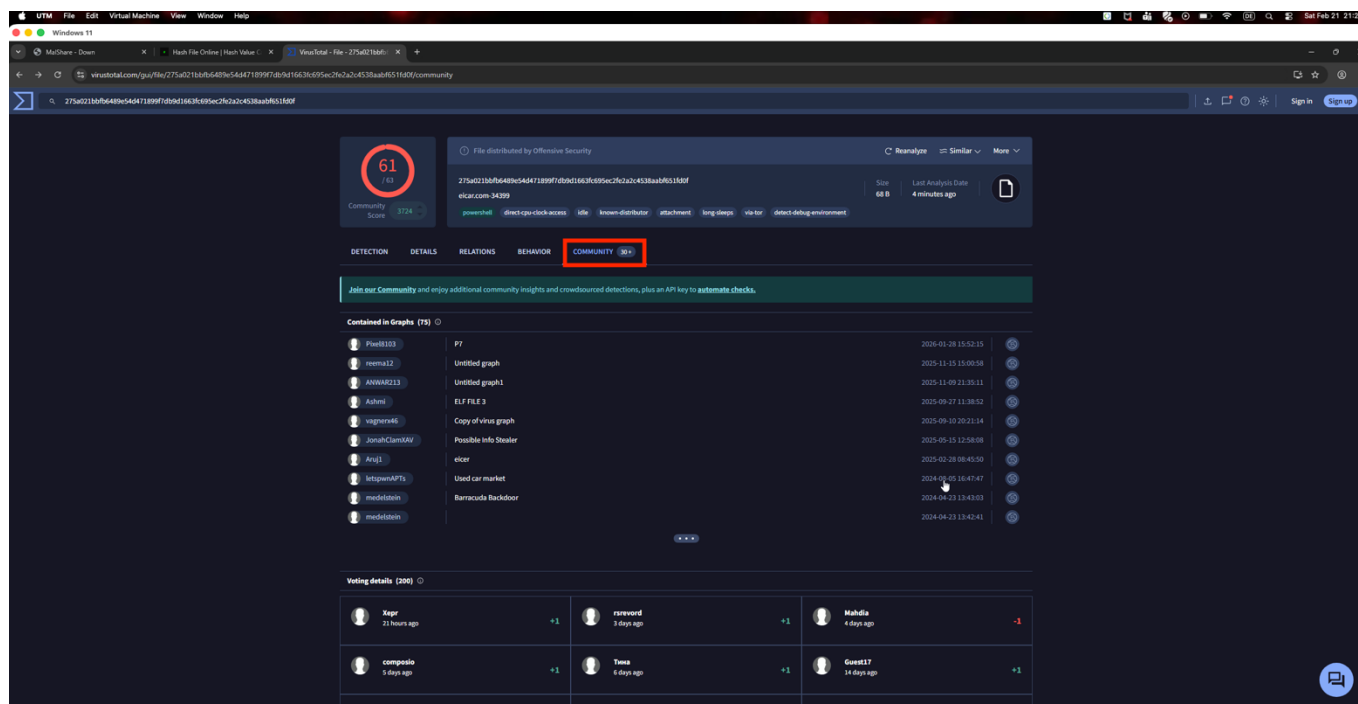


Figure 9: Community Section

This Community Section gives us an overview of vendors or threat hunters behind the creation of the EICAR signature file.

PART B: Reviewing an IP Address with AbuseIPDB

Step 1

- In the address bar, I typed in <https://www.abuseipdb.com/> and verified the captcha

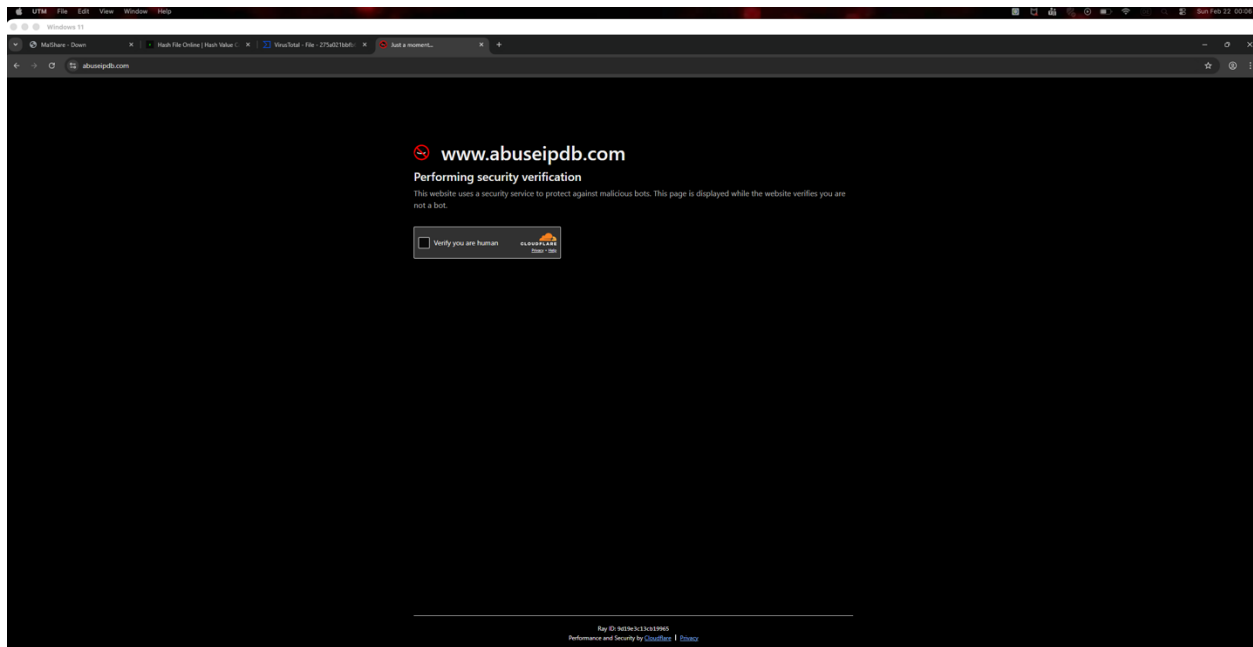


Figure 10.1: Homepage

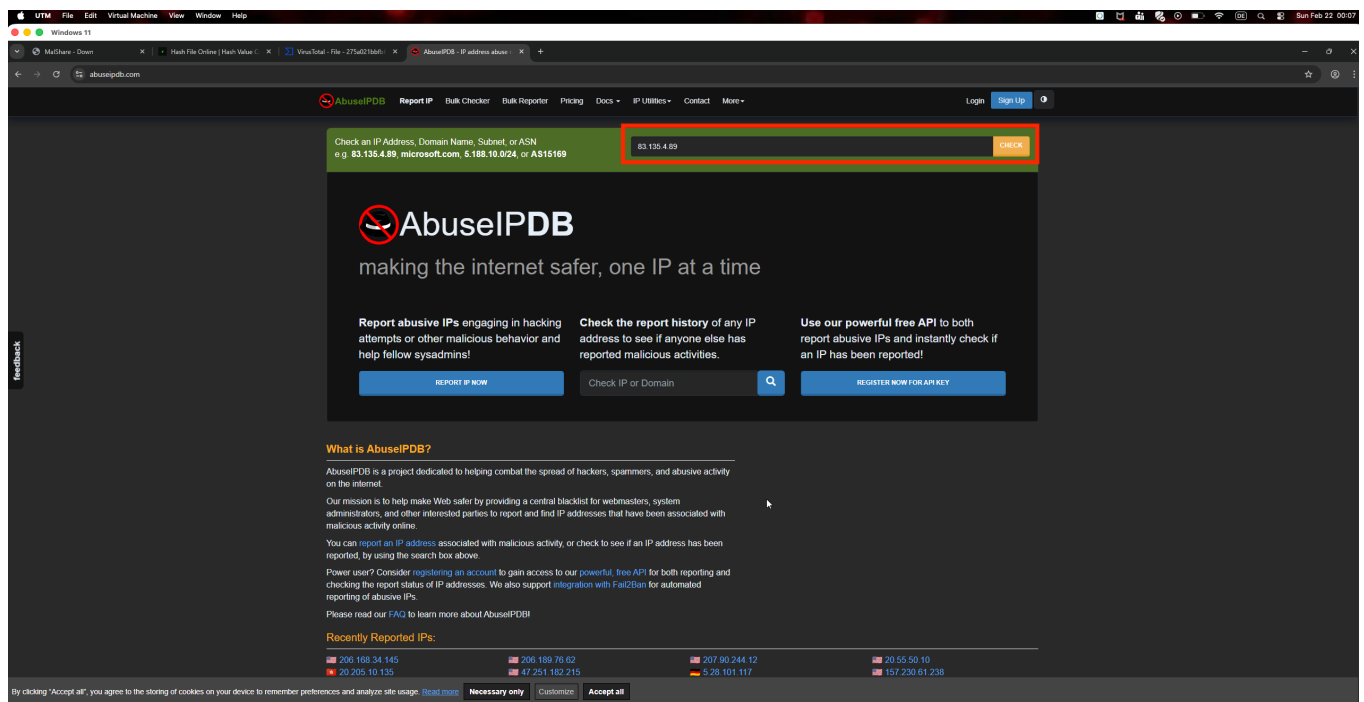


Figure 10.2: Homepage

- In the Search Bar, I typed in **185.220.101.1** IP address

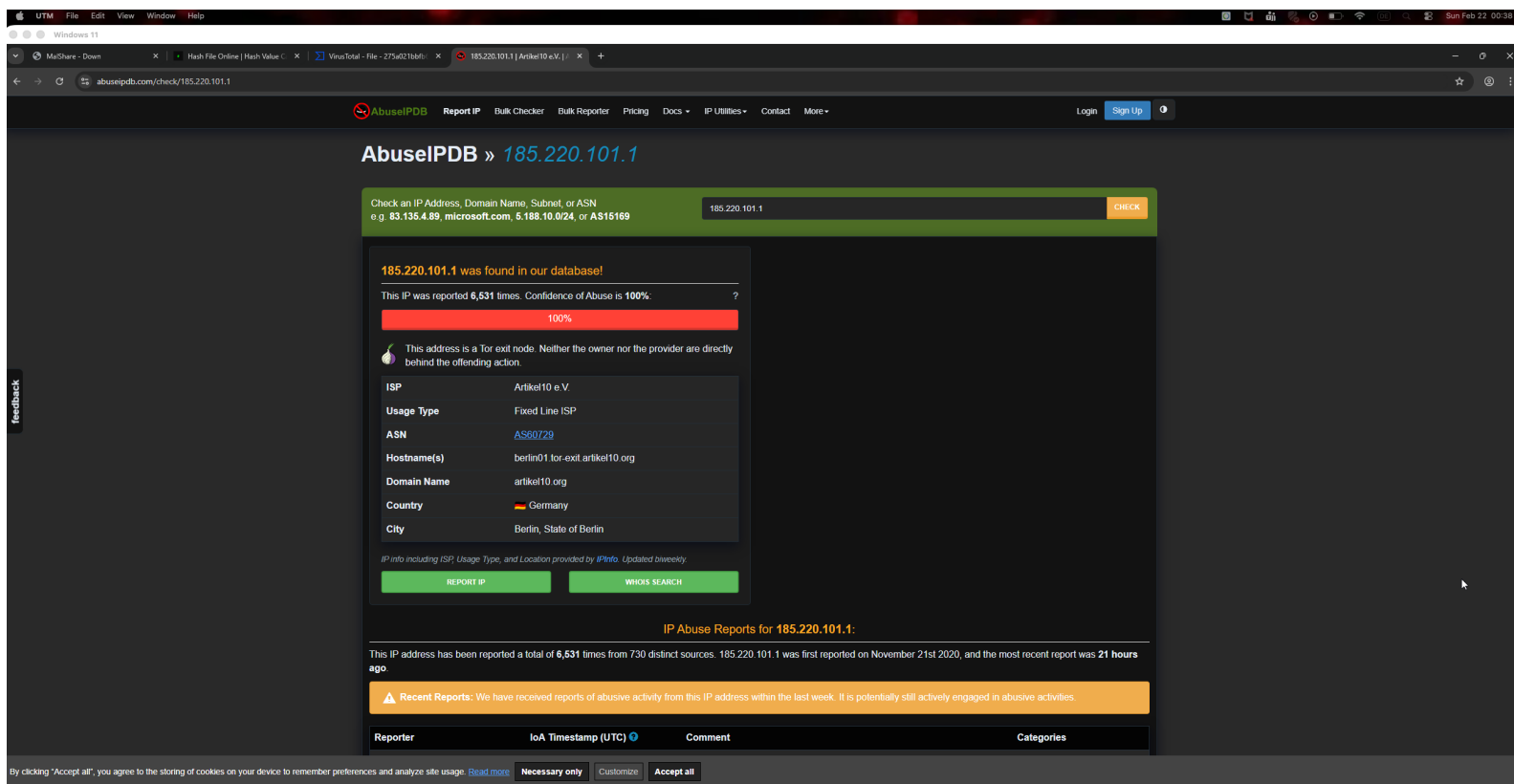


Figure 11: The AbuseIPDB Search Field

- Reviewing the IP details, including the number of abuse reports, categories of abuse (such as hacking, spam), last reported date, and country of origin, the results returned in the search shows show that the IP address was found in the database.
- The IP address has been reported 6,531 times from 730 distinct sources and the confidence of Abuse is rated 100%.
- Abusive activity associated with the IP address was first reported on November 21, 2020 and the abusive activities from the IP address are still ongoing.

PART C: Reviewing a Domain with Cisco Talos Intelligence

- Given the fact that, threat hunting involves proactively searching for malicious events and activities within an environment to uncover ongoing cyberattacks, I believe that every Security Analyst must depend on communications with intelligence groups, creating the room to respond quickly and effectively to mitigate threats.

- Therefore, I decided to explore Cisco Talos Intelligence platform to investigate suspicious domain domains as Ioc.

Step 1

- I opened a new tab

Step 2

- In the address bar, I typed in <https://talosintelligence.com/>

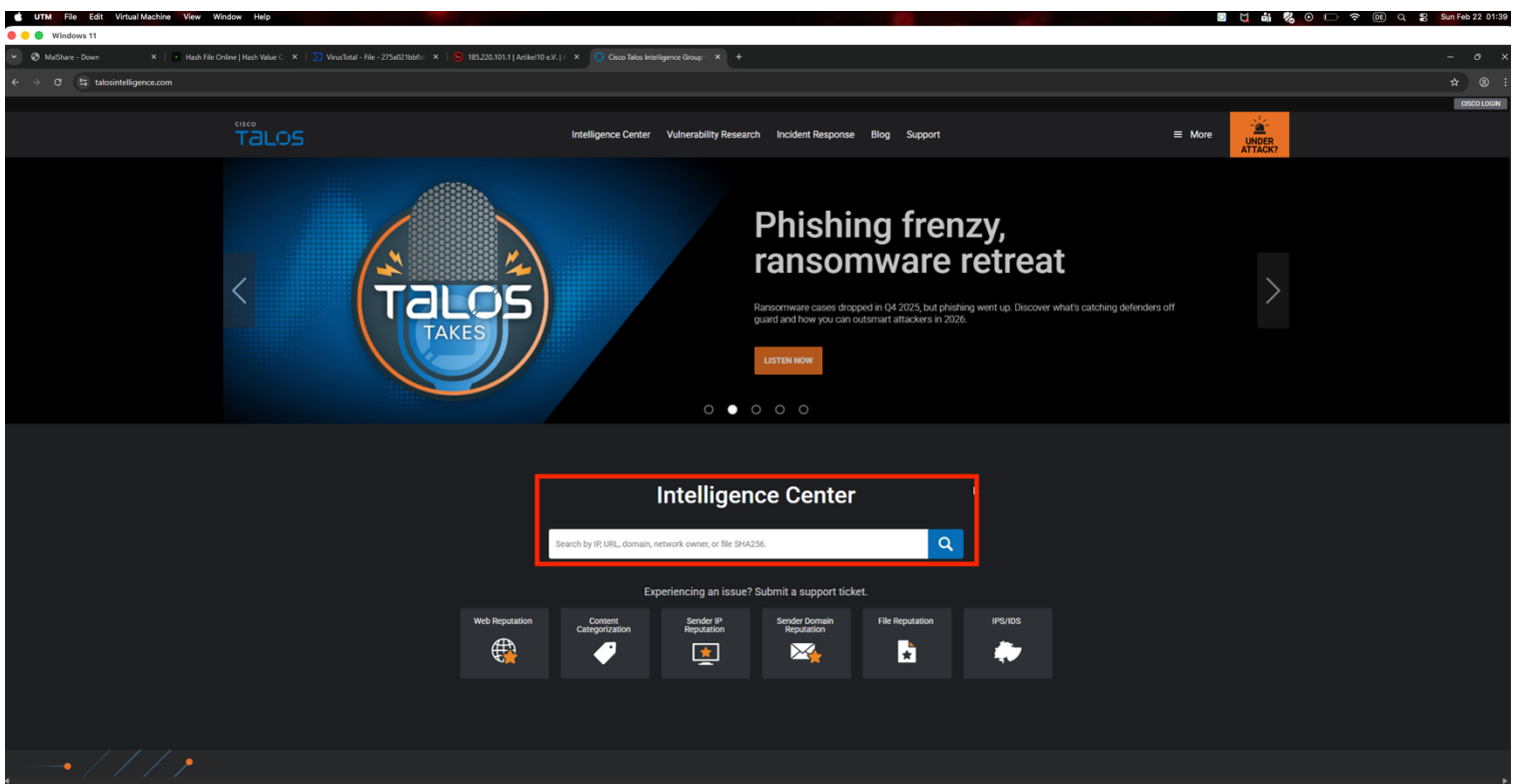


Figure 12: Homepage

Step 3

- In the search field, I entered a suspicious domain as www.example.com and pressed Enter

OWNER DETAILS

DOMAIN: example.com
 HOSTNAME: www.example.com
 NETWORK OWNER: G-CORE LABS S.A.

REPUTATION DETAILS

WEB REPUTATION: Favorable
 EMAIL VOLUME DATA:
 EMAIL VOLUME: 2.8
 VOLUME CHANGE: 0%
 BLOCK LISTS: TALOS SECURITY INTELLIGENCE BLOCK LIST
 ADDED TO BLOCK LIST: No

CONTENT DETAILS

CONTENT CATEGORY: Reference
 Submit Content Categorization Ticket

ADDITIONAL INFORMATION

IP ADDRESSES | WHOIS | EMAIL VOLUME HISTORY | TOP NETWORK OWNERS

Top IP Addresses used to send emails in www.example.com

IP ADDRESS	HOSTNAME	FWO/REV DNS MATCH	LAST DAY VOL	LAST MONTH VOL	BLOCK LISTS	EMAIL REP
185.217.190.0	www.example.com	No	0.0	1.0	0	Poor
180.149.44.17	www.example.com	No	0.0	1.0	0	Poor
186.185.248.8	www.example.com	No	2.4	1.8	0	Neutral
139.20.4.59	www.example.com	No	0.0	1.5	0	Neutral
95.164.61.50	www.example.com	No	0.0	1.0	0	Neutral
89.44.757.163	www.example.com	No	0.0	1.0	0	Neutral
89.43.111.114	www.example.com	No	0.0	1.3	0	Neutral
89.43.105.28	www.example.com	No	0.0	1.3	0	Neutral
43.82.246.31	www.example.com	No	0.0	1.0	0	Neutral
43.82.246.24	www.example.com	No	0.0	1.0	0	Neutral
43.80.210.19	www.example.com	No	0.0	1.5	0	Neutral
43.13.167.109	www.example.com	No	0.0	1.3	0	Neutral
43.19.146.145	www.example.com	No	0.0	1.0	0	Neutral
5.188.6.227	www.example.com	No	0.0	1.0	0	Poor
5.188.6.77	www.example.com	No	0.0	1.3	0	Poor
5.181.75.65	www.example.com	No	0.0	1.3	0	Neutral

Figure 13: Returned Results

The returned results showed several IP addresses associated with Hostname. The email reputation is rated poor.

Results

I have learned the following:

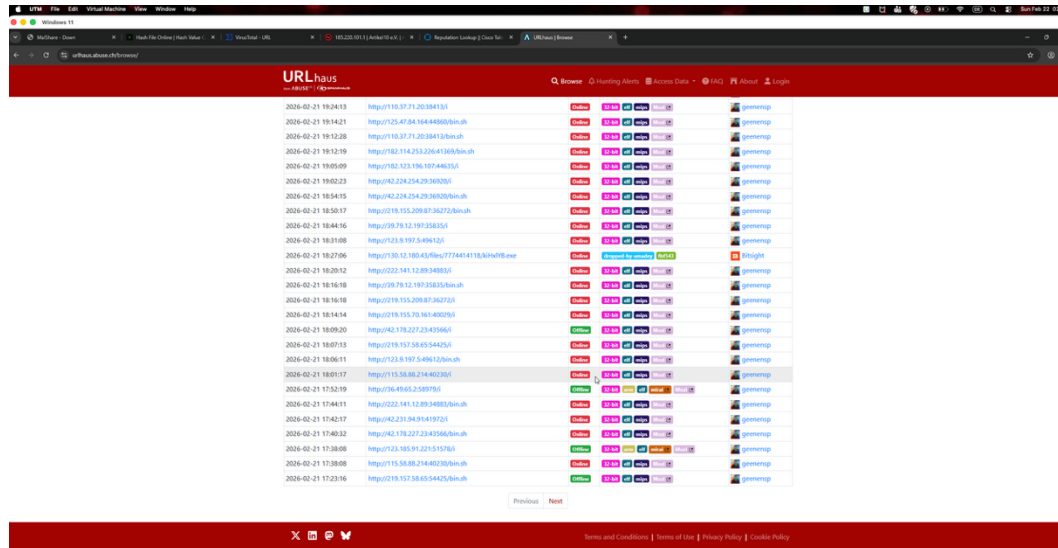
- Detected and reviewed IoC as file hash, ip address, and domain name.
- Reviewed a file hash using VirusTotal, to determine file type, size, detection results, and associated malware family. The use of MalShare was unsuccessful due to an indefinite suspension of the platform after losing their host provider.
- Assessed an IP address through AbuseIPDB to evaluate its reputation, number of abuse reports, categories of abuse, and origin.
- Investigated a domain using Cisco Talos Intelligence to examine reputation score, web category, related IPs, historical data, and security alerts.

Lab Summary

Now, I am equipped with the knowledge and skills to review IoC and threat intelligence sources through this project.

Further Research

Malicious URLs can be obtained from urlhaus.abuse.ch/ for further investigation and analysis. The database contains dozens for malicious URLs for testing purpose.



The screenshot shows the URLhaus website interface. It features a red header with the URLhaus logo and navigation links. The main content area displays a table of malicious URLs. Each row includes a timestamp, a URL, a hash, and a status indicator. The table is sorted by timestamp, showing entries from 2026-02-21 19:24:13 to 2026-02-21 17:23:46. The status indicators include 'Active', 'Inactive', and 'Malware'. The footer contains social media links and a footer bar with 'Previous' and 'Next' buttons.

Timestamp	URL	Hash	Status
2026-02-21 19:24:13	http://10.37.71.2038413/	Active	Malware
2026-02-21 19:14:21	http://125.47.84.16444860/bin.ah	Active	Malware
2026-02-21 19:12:28	http://10.37.71.2038413/bin.ah	Active	Malware
2026-02-21 19:12:19	http://182.114.252.22641369/bin.ah	Active	Malware
2026-02-21 19:05:09	http://182.122.136.10744635/	Active	Malware
2026-02-21 19:02:23	http://42.224.224.2936026/	Active	Malware
2026-02-21 18:54:15	http://42.224.224.2936026/bin.ah	Active	Malware
2026-02-21 18:50:17	http://118.115.2098730272/bin.ah	Active	Malware
2026-02-21 18:44:16	http://198.79.12.19735825/	Active	Malware
2026-02-21 18:31:08	http://123.8.197.5496125/	Active	Malware
2026-02-21 18:27:06	http://180.32.180.454567774414118/bin.ah	Active	Malware
2026-02-21 18:20:42	http://222.345.12.8934883/	Active	Malware
2026-02-21 18:16:18	http://198.79.12.19735825/bin.ah	Active	Malware
2026-02-21 18:16:18	http://118.115.2098730272/	Active	Malware
2026-02-21 18:14:14	http://118.115.2098730272/	Active	Malware
2026-02-21 18:09:20	http://42.178.227.2343566/	Active	Malware
2026-02-21 18:07:13	http://118.115.2098730272/	Active	Malware
2026-02-21 18:06:11	http://123.8.197.5496125/bin.ah	Active	Malware
2026-02-21 18:01:17	http://115.58.86.21440236/	Active	Malware
2026-02-21 17:52:19	http://118.49.65.250976/	Active	Malware
2026-02-21 17:44:11	http://222.345.12.8934883/bin.ah	Active	Malware
2026-02-21 17:40:17	http://42.224.224.2936026/	Active	Malware
2026-02-21 17:40:12	http://42.178.227.2343566/bin.ah	Active	Malware
2026-02-21 17:38:08	http://123.8.197.5496125/	Active	Malware
2026-02-21 17:38:08	http://115.58.86.21440236/bin.ah	Active	Malware
2026-02-21 17:23:46	http://118.115.2098730272/	Active	Malware

Figure 13: Returned Results

Upcoming Additional Tasks:

- Find a suspicious URL or phishing link online (ensure it's legal and ethical to analyze) and then submit the URL to VirusTotal and interpret the scan results.
- Generate hash values (MD5, SHA-1, and SHA-256) for files using various tools and then submit these hash values to VirusTotal and observe how the platform associates different file instances with the same hash.
- Identify a file that is flagged as a false positive by an antivirus engine on VirusTotal.
- Identify patterns or changes in the behavior of a file or URL over time.